

RAA-Gatekeeper

The Read-Only Forensic Auditor for the AI-Agent Era.



Overview

RAA-Gatekeeper is a specialized security middleware designed to solve the "Trust Gap" between local codebases and AI Agents. Built on the **Restrictive Ability (RAA)** protocol, it serves as a **Read-Only Auditor**, ensuring that no code, terminal command, or archive is passed to an LLM without a forensic safety certification.

Prerequisites & Dependencies

Before building the project, ensure the environment meets the following requirements:

1. System Requirements

- **Rust Toolchain:** Install Rust (<https://rust-lang.org/tools/install>) (stable version).
- **Node.js:** Install Node.js (<https://nodejs.org>) (v18 or higher recommended).
- **Package Manager:** npm, yarn, or pnpm.
- **Tauri Dependencies:** Follow the [Tauri Setup Guide \(https://tauri.app/v1/guides/getting-started/prerequisites\)](https://tauri.app/v1/guides/getting-started/prerequisites) for your OS (Windows C++ Build Tools, Linux libwebkit2gtk, etc.).

2. Recommended IDE Setup

- [VS Code \(https://visualstudio.com\)](https://visualstudio.com)
 - **Extensions:**
 - [Svelte for VS Code \(https://visualstudio.com\)](https://visualstudio.com)
 - [Tauri Extension \(https://visualstudio.com\)](https://visualstudio.com)
 - [rust-analyzer \(https://visualstudio.com\)](https://visualstudio.com)
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☐ Getting Started Installation

1. Clone the repository:

```
git clone https://github.com.git  
cd RAA-Gatekeeper
```

2. Install frontend dependencies:

```
npm install
```

Development Mode

Run the application in a live development window:

```
npm run tauri dev
```

Building for Production

Generate a bundled executable (MSI, AppImage, or DMG):

```
npm run tauri build
```

☐ Technical Architecture

- **Frontend:** Svelte 5 / SvelteKit (UI Logic & State Management).
- **Backend:** Rust / Tauri v2 (Forensic Engine & OS Integration).
- **Parallelism:** `Rayon` for multi-core SHA-256 file hashing.
- **Intelligence:** Integrated with xAI Grok-1.5 / Google Gemini for heuristic auditing.

The RAA Mandate

1. **READ-ONLY:** The Gatekeeper **never** modifies user files.
 2. **ADVISORY ONLY:** The app provides forensic diagnostics; the human pilot remains the decision-maker.
 3. **AUDITABLE:** Every interaction is recorded in a `.raa` forensic ledger.
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Project Structure

- `src/` - SvelteKit frontend routes and components.
 - `src-tauri/` - Rust backend, commands (`lib.rs`), and configuration.
 - `raa-test/` - Isolated directory for generated forensic ledgers.
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License

Proprietary / RAA-Certified.

"Trust, then Certify."